

Dataset Validation & EDA Report

Machine Learning-Based Network Intrusion Detection System • Compiled at 2026-07-24 21:27:02

1. Executive Summary & Base Dimensions

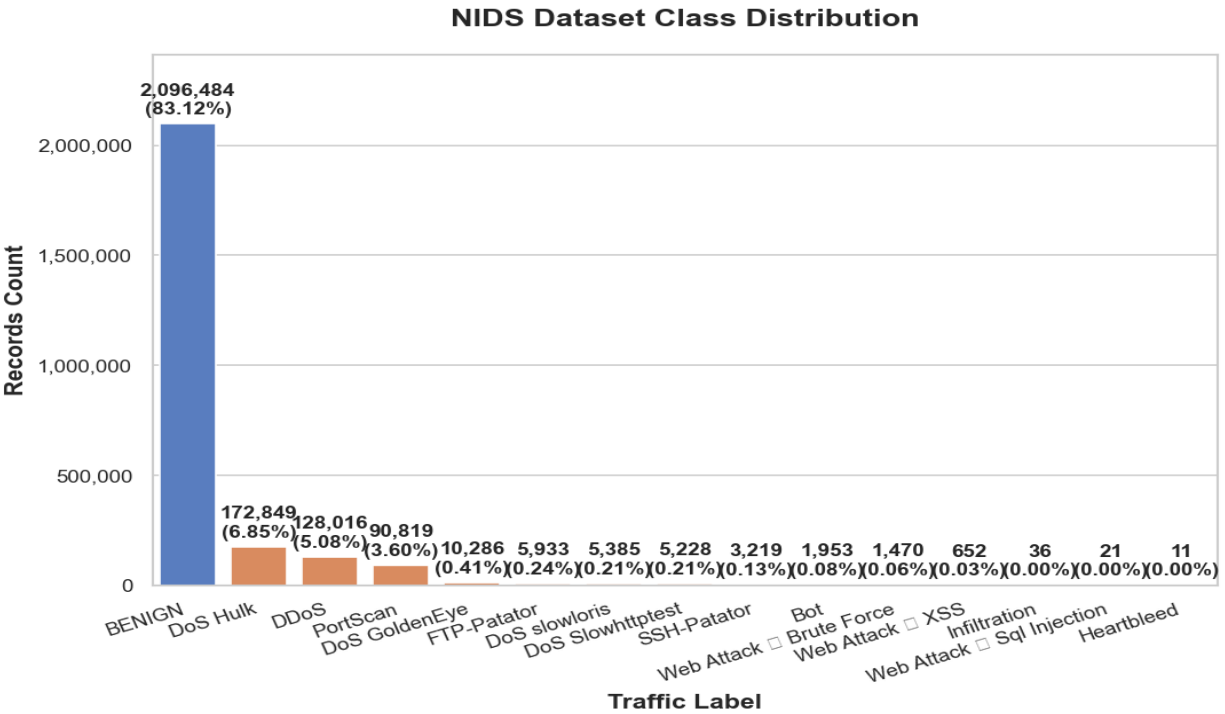
This report outlines the structural parameters and data quality profile of the loaded network telemetry flow records. Automated validation and exploratory analyses were run to prepare the features for machine learning pipelines.

Quality Diagnostic Metric	Value Detail
Total Dataset Rows	2,522,362
Total Dataset Columns	79
Duplicate Rows Count	0
Duplicate Rows Percentage	0.00%
Missing Value Entries Count	0
Overall Null Percentage	0.0000%
Empty Columns Count	0
Constant Columns Count (0 Variance)	8
Infinite Values Count	0
Memory Usage Size	1653.22 MB

2. Target Variable & Class Distribution

Details the frequency breakdown and relative ratios of Benign vs Attack category labels.

Traffic Category Label	Record Count	Percentage
BENIGN	2,096,484	83.116%
DoS Hulk	172,849	6.853%
DDoS	128,016	5.075%
PortScan	90,819	3.601%
DoS GoldenEye	10,286	0.408%
FTP-Patator	5,933	0.235%
DoS slowloris	5,385	0.213%
DoS Slowhttptest	5,228	0.207%
SSH-Patator	3,219	0.128%
Bot	1,953	0.077%
Web Attack ■ Brute Force	1,470	0.058%
Web Attack ■ XSS	652	0.026%
Infiltration	36	0.001%
Web Attack ■ Sql Injection	21	0.001%
Heartbleed	11	0.000%



3. Missing Values Heatmap Grid Analysis

Shows the missing value distributions across top variables containing null values.



Empty Columns Identified:

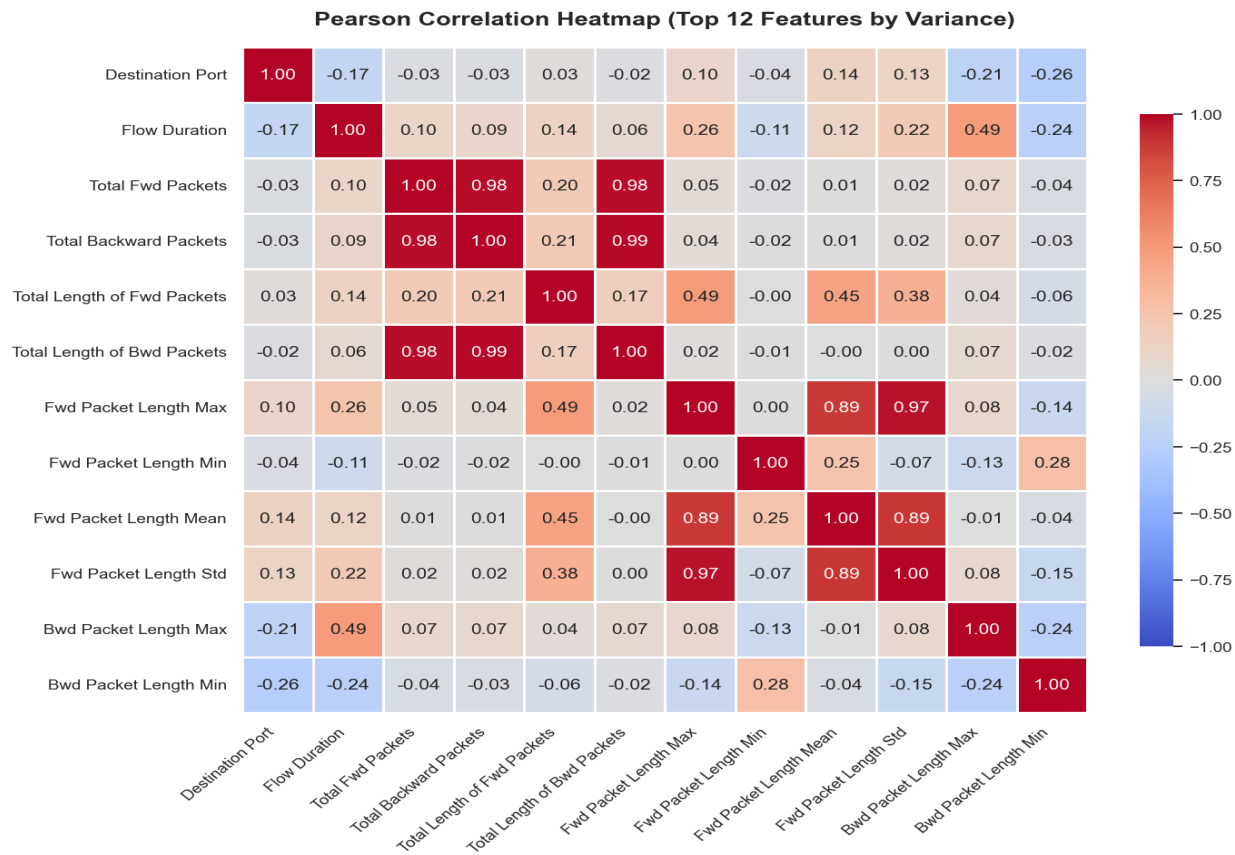
None. All columns contain at least some non-null values.

Constant Columns Identified (0 Variance):

Bwd PSH Flags, Bwd URG Flags, Fwd Avg Bytes/Bulk, Fwd Avg Packets/Bulk, Fwd Avg Bulk Rate, Bwd Avg Bytes/Bulk, Bwd Avg Packets/Bulk, Bwd Avg Bulk Rate

4. Heatmap Correlation Matrix Diagnostics

Presents the Pearson correlation heatmap of features with highest variance to analyze multicollinearity.

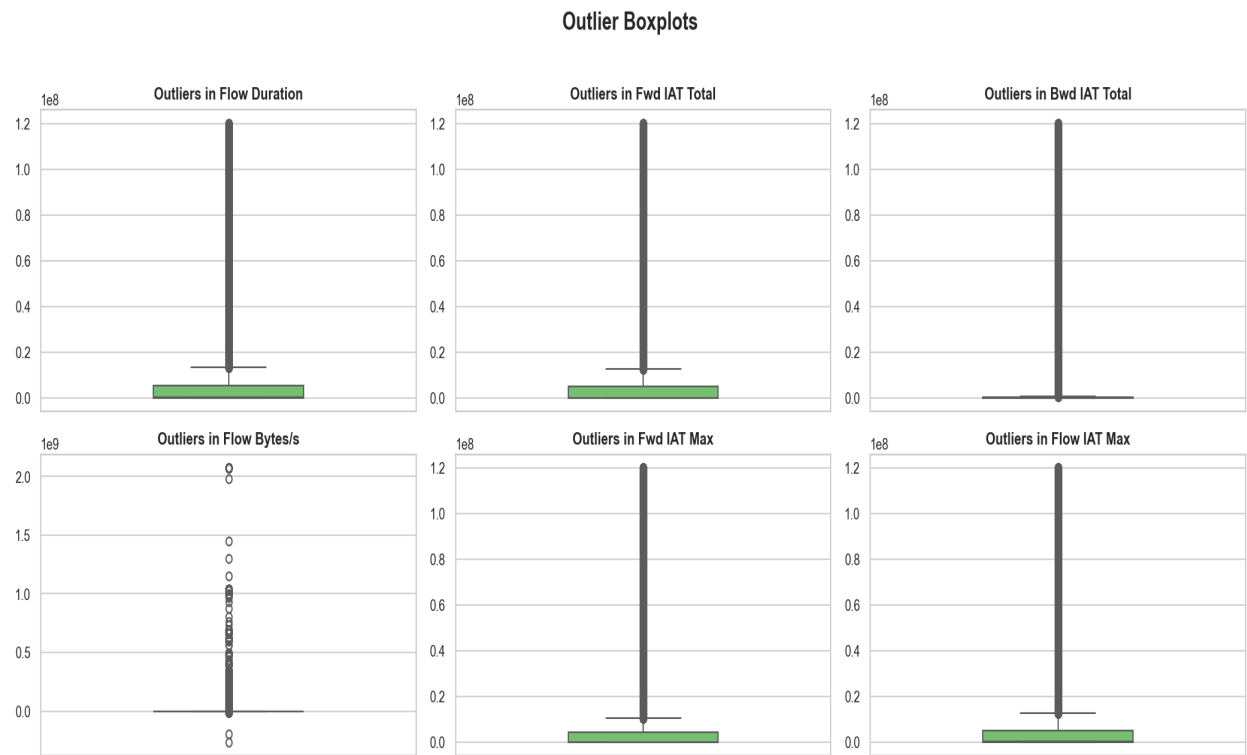


Highly Correlated Feature Pairs (Top 5):

- Fwd PSH Flags <=> SYN Flag Count : Pearson r = 1.0000
- Fwd URG Flags <=> CWE Flag Count : Pearson r = 1.0000
- RST Flag Count <=> ECE Flag Count : Pearson r = 1.0000
- Fwd Packet Length Mean <=> Avg Fwd Segment Size : Pearson r = 1.0000
- Bwd Packet Length Mean <=> Avg Bwd Segment Size : Pearson r = 1.0000

5. Outlier Distributions (Interquartile Range)

Boxplots mapping distributions and outlier ranges for high-variance numerical features.



Outliers Statistics Table (Top 5):

Feature Name	Outliers Count	Percentage
Fwd IAT Std	596,934	23.666%
Bwd Packets/s	590,360	23.405%
Total Length of Bwd Packets	575,636	22.821%
Subflow Bwd Bytes	575,636	22.821%
Idle Mean	567,235	22.488%